ENSIAS

Architecture Big data & applications

Compte rendu

Lab 4: Initiation à MongoDB

Réaliser par :

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Encadrée par :

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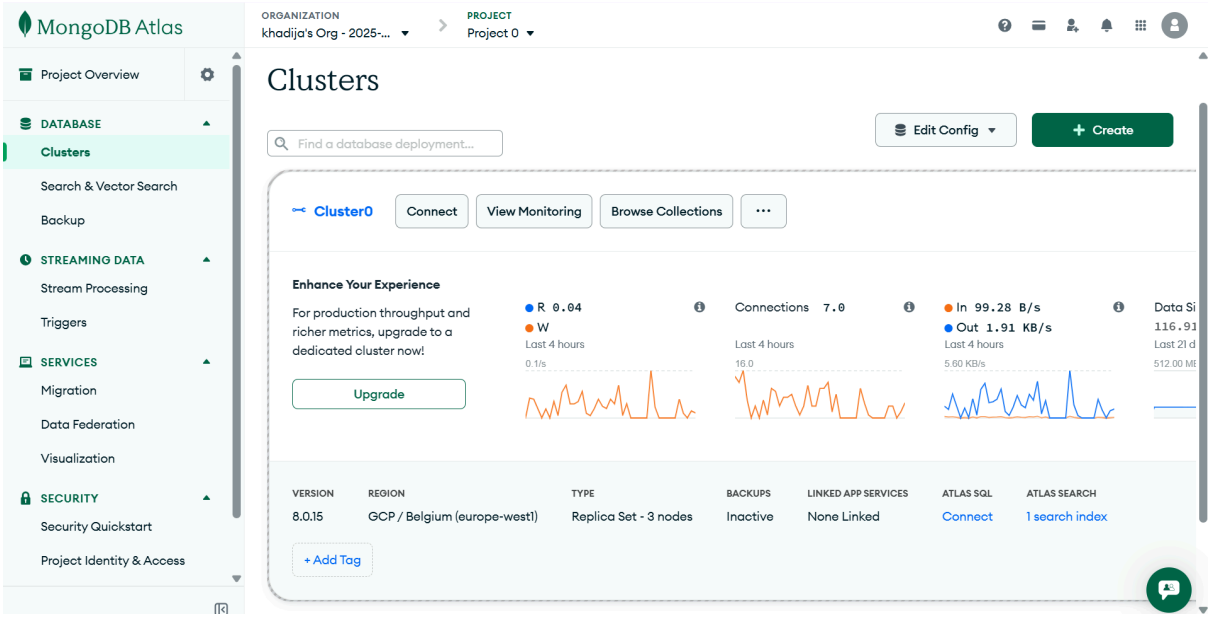
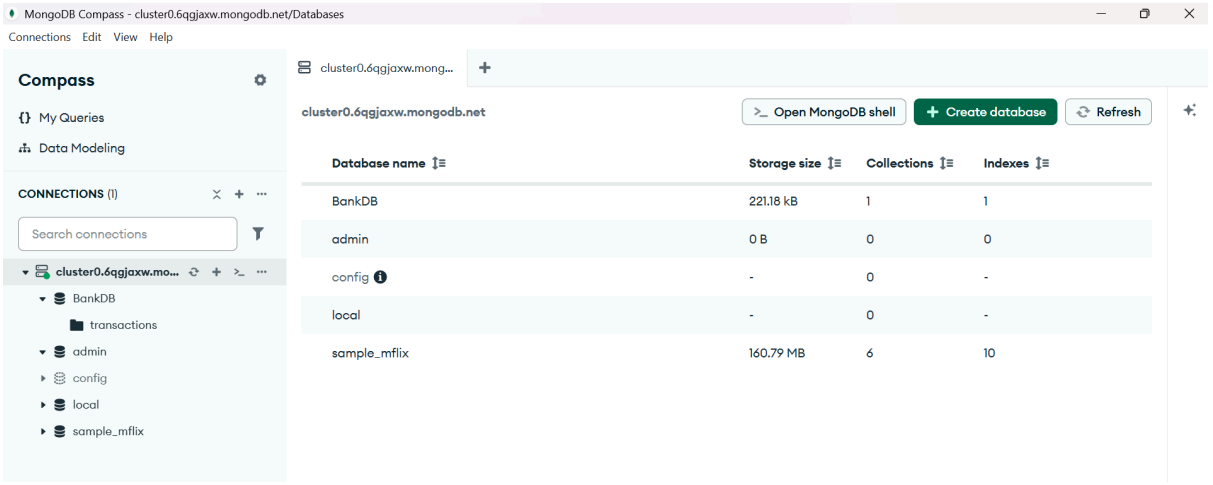
Année scolaire : 2025-2026

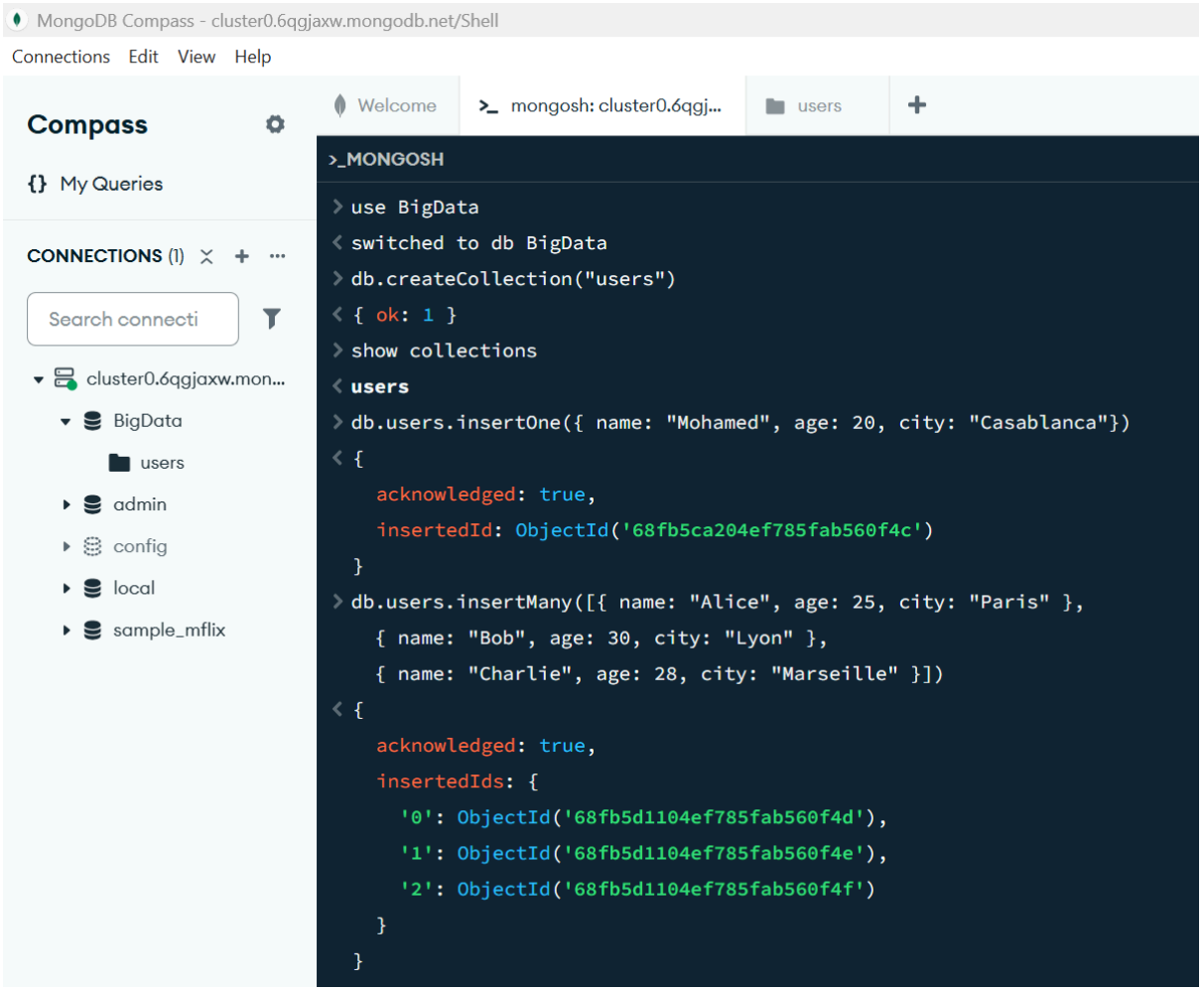
1. Création d’un compte et déploiement d’un cluster :



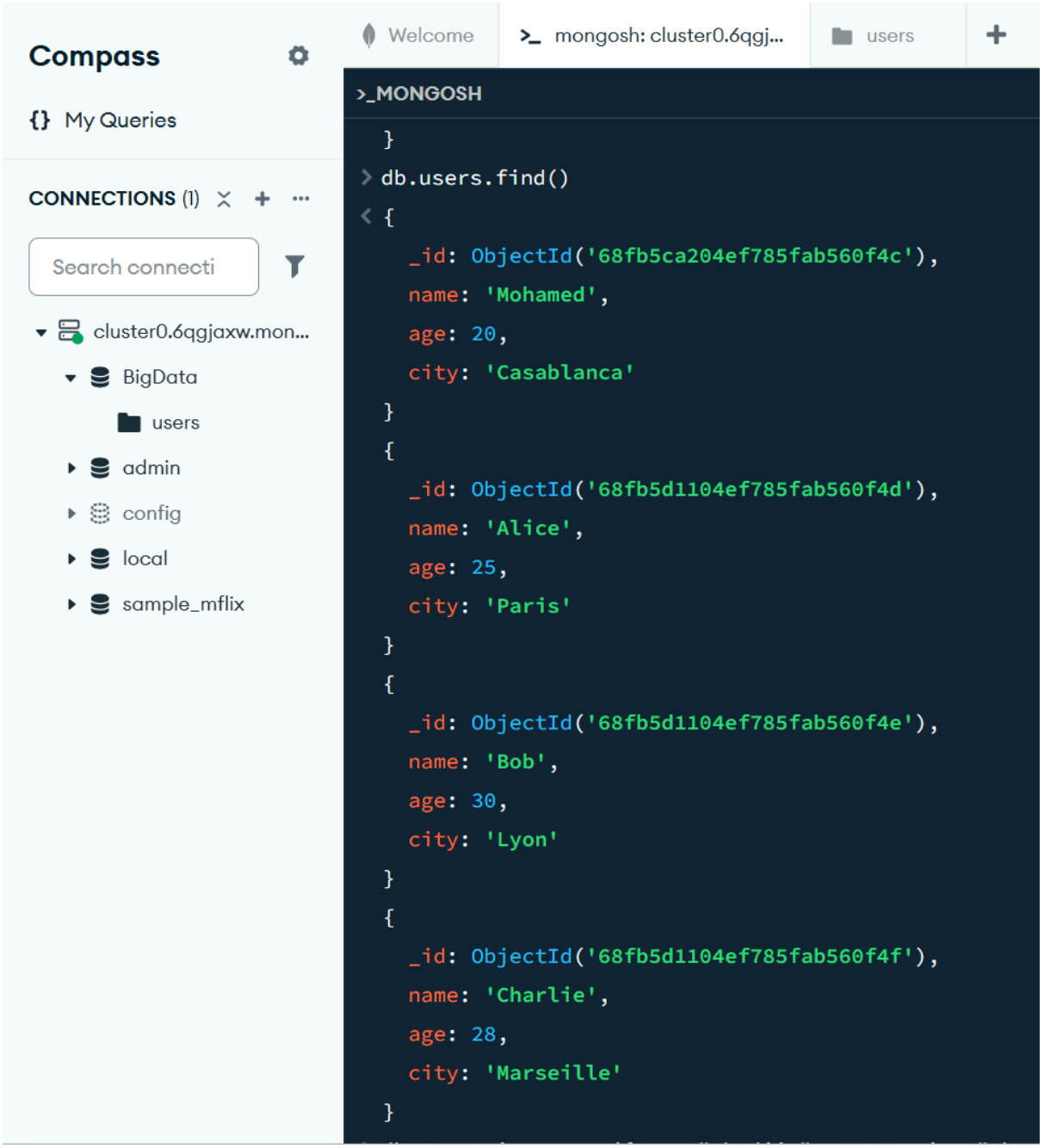
2. Connexion à MongoDB Atlas avec MongoDB Compass :



3. Création d'une base de données et d'une collection:



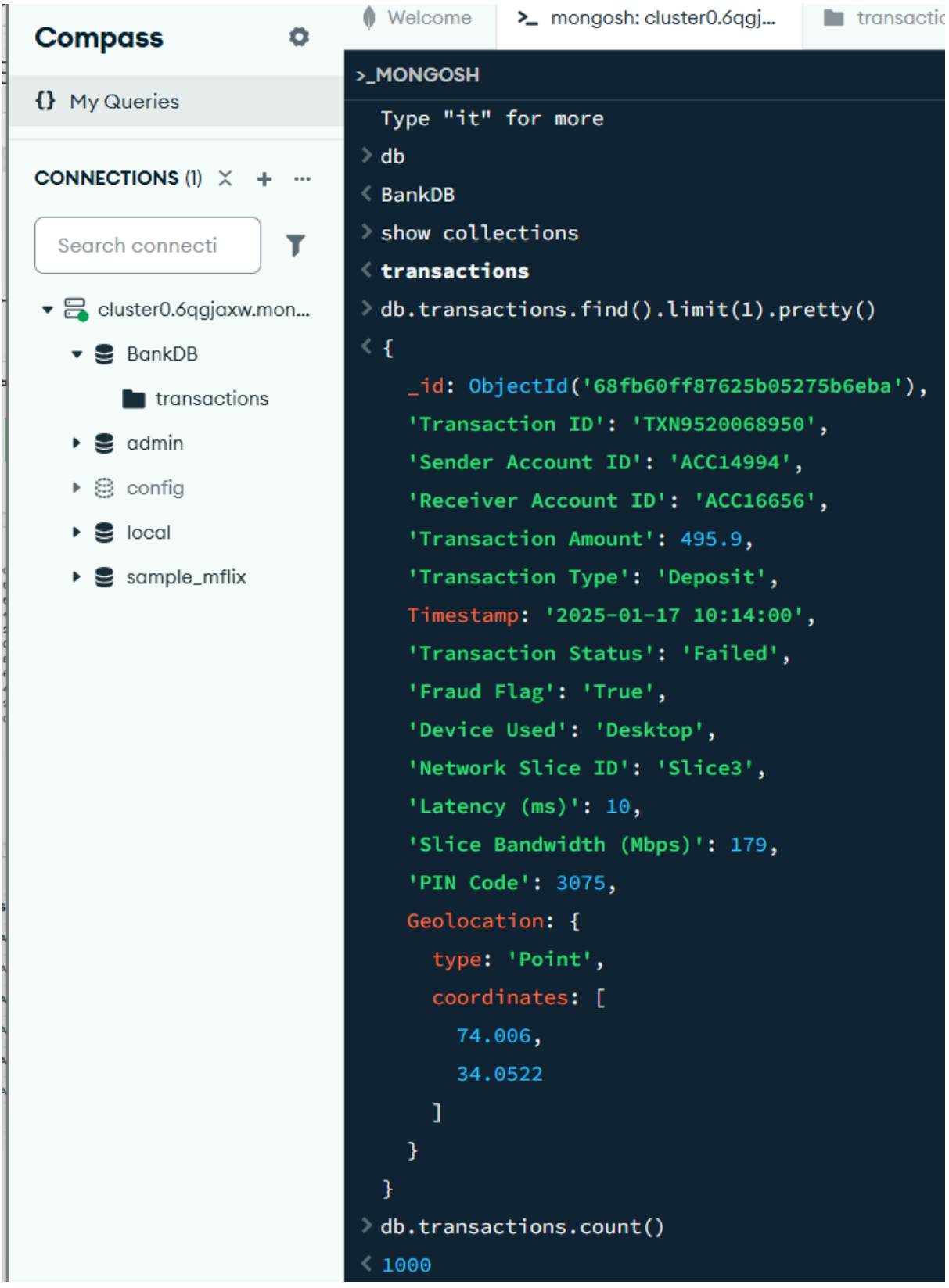
4. Requêtes pour récupérer des données



```
>_MONGOSH
  }
> db.users.insertOne({name:"Khadija",age:23,city:"Khouribga"})
< {
    acknowledged: true,
    insertedId: ObjectId('68fb5d8d04ef785fab560f50')
  }
> db.users.count()
< DeprecationWarning: Collection.count() is deprecated. Use countDocuments or estimatedDocumentCount.
< 5
> db.users.findOne({name:"Alice"})
< {
    _id: ObjectId('68fb5d1104ef785fab560f4d'),
    name: 'Alice',
    age: 25,
    city: 'Paris'
  }
> db.users.find({age:{$gt:20}})
< {
    _id: ObjectId('68fb5d1104ef785fab560f4d'),
    name: 'Alice',
    age: 25,
    city: 'Paris'
  }
  {
    _id: ObjectId('68fb5d1104ef785fab560f4e'),
    name: 'Bob',
    age: 30,
    city: 'Lyon'
  }
  {
    _id: ObjectId('68fb5d1104ef785fab560f4f'),
    name: 'Charlie',
```

```
>_MONGOSH
    age: 28,
    city: 'Marseille'
  }
  {
    _id: ObjectId('68fb5d8d04ef785fab560f50'),
    name: 'Khadija',
    age: 23,
    city: 'Khouribga'
  }
> db.users.updateOne({name:"Alice"},{$set:{age:26}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.users.findOne({name:"Alice"})
< {
    _id: ObjectId('68fb5d1104ef785fab560f4d'),
    name: 'Alice',
    age: 26,
    city: 'Paris'
  }
> db.users.deleteOne({name:"Charlie"})
< {
    acknowledged: true,
    deletedCount: 1
  }
> db.users.drop()
< true
```

5. Importation de fichier dans une collection



```
>_MONGOSH
> db.transactions.aggregate([{$group:{_id:"$Transaction Status",total:{$count:{}}}}])
< {
    _id: 'Success',
    total: 487
  }
  {
    _id: 'Failed',
    total: 513
  }
> db.transactions.aggregate([{$group:{_id:"$Transaction Type",avgAmount:{$avg:"$Transaction Amount"}}}])
< {
    _id: 'Deposit',
    avgAmount: 797.6032278481013
  }
  {
    _id: 'Withdrawal',
    avgAmount: 733.3745806451612
  }
  {
    _id: 'Transfer',
    avgAmount: 780.1512032085561
  }
>  db.transactions.aggregate([
    { $group: { _id: "$Sender Account ID", totalSent: { $sum: "$Transaction Amount" } } },
  {  $sort: { totalSent: -1 } },
    { $limit: 5 }
   ])
< {
    _id: 'ACC37810',
    totalSent: 2757.77
  }
```

```
> db.transactions.aggregate([
    { $group: { _id: "$Sender Account ID", totalSent: { $sum: "$Transaction Amount" } } },
  {  $sort: { totalSent: -1 } },
    { $limit: 5 }
   ])
< {
    _id: 'ACC37810',
    totalSent: 2757.77
  }
  {
    _id: 'ACC75741',
    totalSent: 1918.82
  }
  {
    _id: 'ACC89865',
    totalSent: 1692.55
  }
  {
    _id: 'ACC44804',
    totalSent: 1497.76
  }
  {
    _id: 'ACC67128',
    totalSent: 1495.01
  }
```

```
> db.transactions.aggregate([
    { $group: { _id: "$Transaction Type", total: { $sum: "$Transaction Amount" } } }
  ])
< {
    _id: 'Withdrawal',
    total: 227346.12
  }
  {
    _id: 'Transfer',
    total: 291776.55
  }
  {
    _id: 'Deposit',
    total: 252042.62
  }
```

```
> db.transactions.aggregate([
    { $group: { _id: "$Network Slice ID", avgLatency: { $avg: "$Latency (ms)" } } }
   ])
< {
    _id: 'Slice1',
    avgLatency: 11.91640866873065
  }
  {
    _id: 'Slice3',
    avgLatency: 11.86053412462908
  }
  {
    _id: 'Slice2',
    avgLatency: 11.3
  }
```

```
> db.transactions.aggregate([
    { $match: { "Transaction Status": "Failed" } },
    { $group: { _id: "$Device Used", totalFailed: { $count: {} } } }
  ])
< {
    _id: 'Mobile',
    totalFailed: 270
  }
  {
    _id: 'Desktop',
    totalFailed: 243
  }
```

```
> db.transactions.find({ "Fraud Flag": "True", "Transaction Amount": { $gt: 1000 } })
< {
    _id: ObjectId('68fb60ff87625b05275b6ebd'),
    'Transaction ID': 'TXN2214150284',
    'Sender Account ID': 'ACC48650',
    'Receiver Account ID': 'ACC76457',
    'Transaction Amount': 1129.88,
    'Transaction Type': 'Transfer',
    Timestamp: '2025-01-17 10:56:00',
    'Transaction Status': 'Success',
    'Fraud Flag': 'True',
    'Device Used': 'Mobile',
    'Network Slice ID': 'Slice3',
    'Latency (ms)': 10,
    'Slice Bandwidth (Mbps)': 127,
    'PIN Code': 6374,
    Geolocation: {
      type: 'Point',
      coordinates: [
        74.006,
        34.0522
      ]
    }
  }
  {
    _id: ObjectId('68fb60ff87625b05275b6ec1'),
    'Transaction ID': 'TXN3109277527',
```

```
> db.transactions.find({"Transaction Amount":{$gte:100,$lte:200}}).count()
< 66
```

```
> db.transactions.aggregate([
    { $group: { _id: null, totalAmount: { $sum: "$Transaction Amount" } } }
  ])
< {
    _id: null,
    totalAmount: 771165.29
  }
```

```
> db.transactions.aggregate([
    { $group: { _id: "$Transaction Status", count: { $count: {} } } }
  ])
< {
    _id: 'Failed',
    count: 513
  }
  {
    _id: 'Success',
    count: 487
  }
```

```
> db.transactions.countDocuments({ "Fraud Flag": "True" })
< 481
```

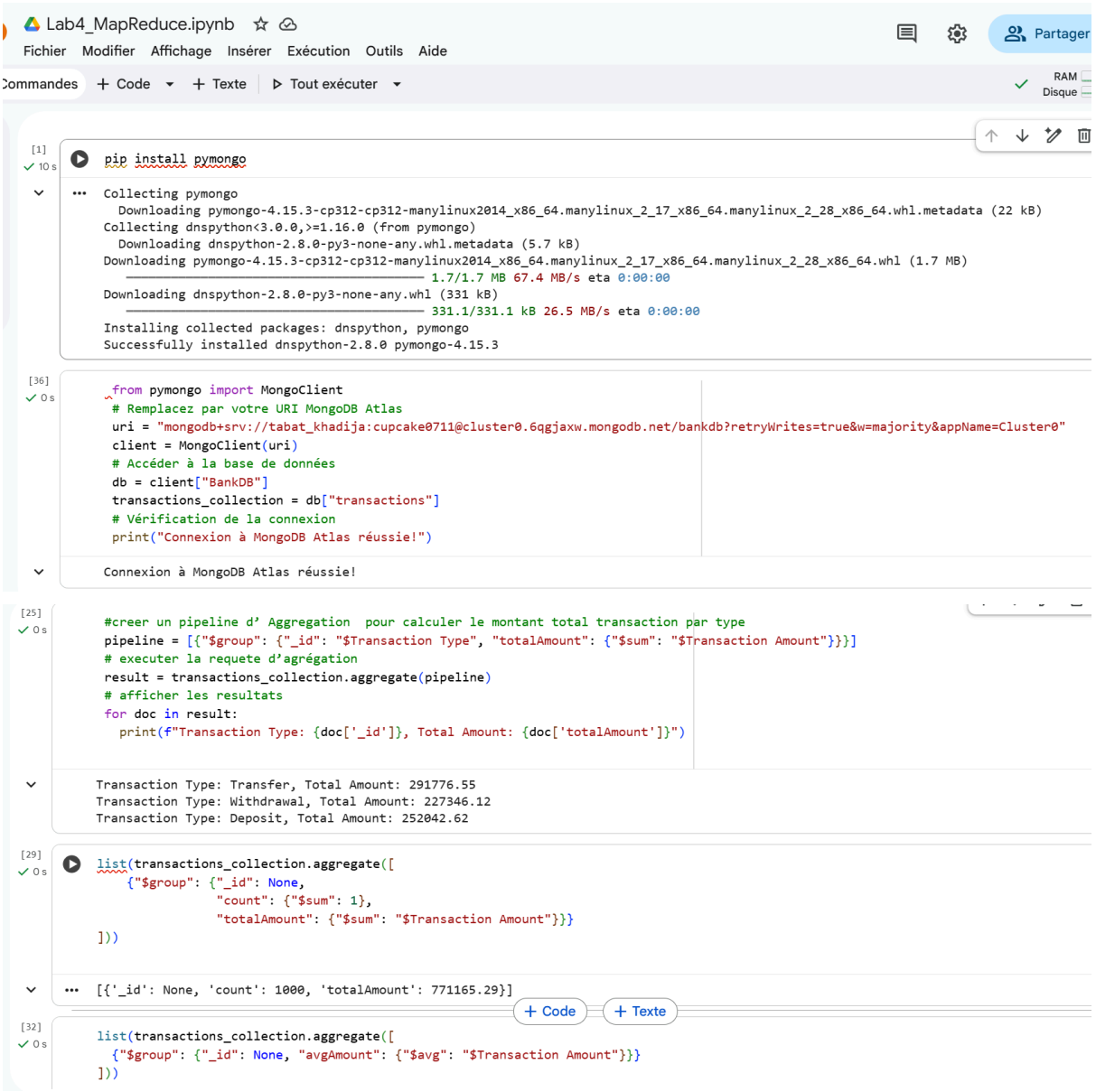
```
> db.transactions.find({ "Fraud Flag": "True", "Transaction Amount": { $gt: 1000 }
< {
    _id: ObjectId('68fb60ff87625b05275b6ebd'),
    'Transaction ID': 'TXN2214150284',
    'Sender Account ID': 'ACC48650',
    'Receiver Account ID': 'ACC76457',
    'Transaction Amount': 1129.88,
    'Transaction Type': 'Transfer',
    Timestamp: '2025-01-17 10:56:00',
    'Transaction Status': 'Success',
    'Fraud Flag': 'True',
    'Device Used': 'Mobile',
    'Network Slice ID': 'Slice3',
    'Latency (ms)': 10,
    'Slice Bandwidth (Mbps)': 127,
    'PIN Code': 6374,
    Geolocation: {
      type: 'Point',
      coordinates: [
        74.006,
        34.0522
      ]
    }
  }
  {
    _id: ObjectId('68fb60ff87625b05275b6ec1'),
    'Transaction ID': 'TXN3109277527',
```

```
> db.transactions.find({"Transaction Status":"Failed"})
< {
    _id: ObjectId('68fb60ff87625b05275b6eba'),
    'Transaction ID': 'TXN9520068950',
    'Sender Account ID': 'ACC14994',
    'Receiver Account ID': 'ACC16656',
    'Transaction Amount': 495.9,
    'Transaction Type': 'Deposit',
    Timestamp: '2025-01-17 10:14:00',
    'Transaction Status': 'Failed',
    'Fraud Flag': 'True',
    'Device Used': 'Desktop',
    'Network Slice ID': 'Slice3',
    'Latency (ms)': 10,
    'Slice Bandwidth (Mbps)': 179,
    'PIN Code': 3075,
    Geolocation: {
      type: 'Point',
      coordinates: [
        74.006,
        34.0522
      ]
    }
  }
  {
    _id: ObjectId('68fb60ff87625b05275b6ebc'),
    'Transaction ID': 'TXN4407425052',
    'Sender Account ID': 'ACC56321',
    'Receiver Account ID': 'ACC92481',
    'Transaction Amount': 862.47,
    'Transaction Type': 'Withdrawal',
    Timestamp: '2025-01-17 10:50:00',
    'Transaction Status': 'Failed',
    'Fraud Flag': 'False',
    'Device Used': 'Mobile',
    'Network Slice ID': 'Slice1',
    'Latency (ms)': 4,
    'Slice Bandwidth (Mbps)': 53,
    'PIN Code': 8039,
    Geolocation: {
```

>_MONGOSH

```
> db.transactions.find({"Fraud Flag": "True"})
< {
    _id: ObjectId('68fb60ff87625b05275b6eba'),
    'Transaction ID': 'TXN9520068950',
    'Sender Account ID': 'ACC14994',
    'Receiver Account ID': 'ACC16656',
    'Transaction Amount': 495.9,
    'Transaction Type': 'Deposit',
    Timestamp: '2025-01-17 10:14:00',
    'Transaction Status': 'Failed',
    'Fraud Flag': 'True',
    'Device Used': 'Desktop',
    'Network Slice ID': 'Slice3',
    'Latency (ms)': 10,
    'Slice Bandwidth (Mbps)': 179,
    'PIN Code': 3075,
    Geolocation: {
      type: 'Point',
      coordinates: [
        74.006,
        34.0522
      ]
    }
  }
  {
    _id: ObjectId('68fb60ff87625b05275b6ebd'),
    'Transaction ID': 'TXN2214150284',
    'Sender Account ID': 'ACC48650',
    'Receiver Account ID': 'ACC76457',
    'Transaction Amount': 1129.88,
    'Transaction Type': 'Transfer',
    Timestamp: '2025-01-17 10:56:00',
    'Transaction Status': 'Success',
    'Fraud Flag': 'True',
    'Device Used': 'Mobile',
```

6. Manipuler MapReduce sur la collection transactions



[29] ✓ 0 s

```
list(transactions_collection.aggregate([
    {"$group": {"_id": None,
                "count": {"$sum": 1},
                "totalAmount": {"$sum": "$Transaction Amount"}}}
]))
```

```
[{'_id': None, 'count': 1000, 'totalAmount': 771165.29}]
```

[32] ✓ 0 s

```
list(transactions_collection.aggregate([
  {"$group": {"_id": None, "avgAmount": {"$avg": "$Transaction Amount"}}}
]))
```

```
[{'_id': None, 'avgAmount': 771.16529}]
```

[33] ✓ 0 s

```
list(transactions_collection.aggregate([
  {"$group": {"_id": None,
              "minAmount": {"$min": "$Transaction Amount"},
              "maxAmount": {"$max": "$Transaction Amount"}}}
]))
```

```
[{'_id': None, 'minAmount': 51.89, 'maxAmount': 1497.76}]
```

[34] ✓ 0 s

```
list(transactions_collection.aggregate([
  {"$group": {"_id": "$Transaction Status", "n": {"$sum": 1}}},
  {"$sort": {"n": -1}}
]))
```

```
[{'_id': 'Failed', 'n': 513}, {'_id': 'Success', 'n': 487}]
```

[35] ✓ 0 s

```
list(transactions_collection.aggregate([
  {"$group": {"_id": "$Transaction Type",
              "sum": {"$sum": "$Transaction Amount"},
              "avg": {"$avg": "$Transaction Amount"}}},
  {"$sort": {"sum": -1}}
]))
```

```
[{'_id': 'Transfer', 'sum': 291776.55, 'avg': 780.1512032085561},
 {'_id': 'Deposit', 'sum': 252042.62, 'avg': 797.6032278481013},
 {'_id': 'Withdrawal', 'sum': 227346.12, 'avg': 733.3745806451612}]
```

7. Dashboard avec MongoDB Atlas